2 Left Cosets and Quotient Groups

In this section whenever we consider a group $(G, \circ)$ we will omit the notation $\circ$, i.e. we will simply write $xy = x \circ y$ if $x, y \in G$.

Definition 2.1 (Subgroup Relation).
Let G be a group and $H \subset G$ a subgroup. We define the relation R among elements of G by $xRy \iff x^{-1}y \in H$.

From the definition, it follows that xRy if and only if there exists some $h \in H$ such that $y = xh$.

Definition.
A relation R on a set S is an *equivalence relation* if:

1. R is *reflexive*: for all $x \in S$, we have xRx.
2. R is *symmetric*: for all $x, y \in S$, if xRy, then yRx.
3. R is *transitive*: for all $x, y, z \in S$, if xRy and yRz, then xRz.

Given an equivalence relation R on a set S and an element x of S, the *equivalence class* of x, denoted $\bar{x}$ is the set $\bar{x} = \{y \in S : xRy\}$.

Lemma 2.1.
The relation R is an equivalence relation. If $x \in G$ then its equivalence class, denoted $\bar{x}$, for the relation R is

$$\bar{x} = xH = \{xh : h \in H\}.$$

Proof. R is reflexive: xRx since $x^{-1}x = e_G \in H$. R is symmetric: assume xRy, i.e. $x^{-1}y \in H$, then $(x^{-1}y)^{-1} = y^{-1}(x^{-1})^{-1} = y^{-1}x \in H$ (since H is a subgroup) which means yRx. R is transitive: assume xRy, i.e. $x^{-1}y \in H$, and yRz, i.e. $y^{-1}z \in H$, then $(x^{-1}y)(y^{-1}z) = x^{-1}yy^{-1}z = x^{-1}z \in H$ (since H is a subgroup) which means xRz.

The equivalence class of $x \in G$ is

$$\bar{x} = \{y \in G : xRy\} = \{y \in G : y = xh, h \in H\} = xH = \{xh : h \in H\}.$$

□

Definition 2.2 (Left Cosets).
The equivalence class $xH = \{xh : h \in H\}$ is called a *left coset* of H; it is a subset of G.

We denote the set of all left cosets of H by

$$(G/H)_{\text{left}} = \{xH : x \in G\}.$$

We usually omit the word left and write G/H instead of $(G/H)_{\text{left}}$.

The cardinality of G/H is denoted $|G : H|$ and is called the *index* of H in G.

As R is an equivalence relation, we have

$$G = \bigcup_{x \in G} \bar{x} = \bigcup_{x \in G} xH$$

where the union is disjoint, i.e. given $x, y \in G$ then either $xH = yH$; which happens if and only if xRy, or $xH \bigcap yH = \emptyset$. The proof of this claim is essentially that given in MTH1001. It generalises to all equivalence relations.

Note that

$$\overline{e_G} = e_G H = \{e_G h : h \in H\} = H \in G/H.$$

If G is finite then G/H is a finite set, its cardinality is denoted

$$|G : H| = |G/H|$$

and is called the *index* of H in G. Clearly $|G : H| \leq |G|$.

Example 2.1.

1. Let $G = (\mathbf{Z}, +)$ and $H = n\mathbf{Z}$ $(n \geq 0)$. In this case

   $$aRb \iff -a + b = b - a \in H = n\mathbf{Z},$$

   i.e. a is congruent to b modulo n. Let $a \in \mathbf{Z}$ with $a = kn + r$, $0 \leq r < n$ (Euclidean division). Then aRr since $-a + r = (-k)n \in n\mathbf{Z}$, and $\bar{a} = \bar{r} = r + n\mathbf{Z}$. Thus

   $$\mathbf{Z}/n\mathbf{Z} = \{\bar{0}, \bar{1}, \cdots, \overline{n-1}\}$$

   is just the set $\mathbf{Z}_n$ of integers modulo n.
2. Let $G = \mathbf{Z}_4 = \{[0]_4, [1]_4, [2]_4, [3]_4\}$ and $H = \langle [2]_4 \rangle = \{[0]_4, [2]_4\}$. In this case

   $$\overline{[0]_4} = \overline{[2]_4} = [0]_4 + H = H$$

   since $[2]_4 - [0]_4 = [2]_4 \in H$, and

   $$\overline{[1]_4} = \overline{[3]_4} = [1]_4 + H = \{[1]_4, [3]_4\}$$

   since $[3]_4 - [1]_4 = [2]_4 \in H$. Here $|G : H| = 2$.
3. Let $D_6 = \{1, r, r^2, s, sr, sr^2\}$ and $H = \langle r \rangle = \{1, r, r^2\}$. Then

   $$\bar{1} = \bar{r} = \overline{r^2} = 1H = H$$

   and

   $$\bar{s} = \overline{sr} = \overline{sr^2} = sH = \{s, sr, sr^2\}$$

   since $s^{-1}(sr) = s(sr) = s^2 r = r \in H$ and $s^{-1}(sr^2) = s(sr^2) = s^2 r^2 = r^2 \in H$. Here $|G : H| = 2$.

Theorem 2.2 (Lagrange's Theorem)**.**
Let G be a finite group and H a subgroup. Then

$$|G| = |H||G/H| = |H||G : H|.$$

Proof. First we show that all left cosets have the same cardinality $|H| = |xH|$. Let $x \in G$ fixed. The map $\sigma : H \to xH$ defined by $\sigma(h) = xh$ is a bijection. Indeed, it is clearly surjective. Assume $\sigma(h_1) = xh_1 = \sigma(h_2) = xh_2$ then $x^{-1}xh_1 = h_1 = x^{-1}xh_2 = h_2$. Since σ is bijective, H and xH have the same cardinality. We know that G is the disjoint union of the left cosets of H and all these cosets have the same cardinality: equal that of H. Thus $|G|$ equals the number of left cosets; i.e. $|G : H|$, times the cardinality of a coset which equals $|H|$. □

Corollary 2.3.

1. *Let G be a finite group and H a subgroup. Then $|H|$ divides $|G|$.*
2. *Let G be a finite group and $x \in G$ then $\mathrm{ord}(x) = |\langle x \rangle|$ divides $|G|$.*

Theorem 2.4 (Cauchy's Theorem).
Let G be a finite group and p a prime number dividing $|G|$. Then $\exists g \in G$ with $\mathrm{ord}(g) = p$, and there exists a subgroup H of G with $|H| = p$.

Definition 2.3 (Normal Subgroup).
A subgroup H of G is called *normal* if

$$xH = \{xh \ : \ h \in H\} = Hx = \{h'x \ : \ h' \in H\}, \quad \forall x \in G.$$

This condition is satisfied if G is abelian since the elements of G commute: $xh = hx$. Thus every subgroup of an abelian group is a normal subgroup.

Example 2.2.

1. Let $D_6 = \{1, r, r^2, s, sr, sr^2\}$ and $H = \langle r \rangle = \{1, r, r^2\}$. Then H has 2 left cosets $\bar{1} = \bar{r} = \overline{r^2} = 1H = H$ and $\bar{s} = \overline{sr} = \overline{sr^2} = sH = \{s, sr, sr^2\}$ (see Examples 2.1). Further H is normal. Indeed $rH = r^2H = Hr = Hr^2 = H$ and $sH = srH = sr^2H = Hs = Hsr = Hsr^2$, since $sr^j = r^{-j}s$ hence
$$sr^k r^j = sr^{k+j} = sr^{j+k} = sr^j r^k = r^{-j} sr^k$$
for $0 \leq j < 3$, and $0 \leq k < 3$.
2. The subgroup $K = \{1, s\} = \langle s \rangle$ of $D_8 = \{1, r, r^2, r^3, s, sr, sr^2, sr^3\}$ is NOT normal. Indeed $rK = \{r, rs = sr^3\} \neq Kr = \{r, sr\}$ as $sr^3 \neq sr$ (since $r^3 \neq r$).
3. If $f : G \to G'$ is a group homomorphism then $\mathrm{Ker}(f)$ is a normal subgroup of G.

   Proof. Let $x \in G$, we show $x\,\mathrm{Ker}(f) = \mathrm{Ker}(f)x$. First we show $x\,\mathrm{Ker}(f) \subset \mathrm{Ker}(f)x$. Let $h \in \mathrm{Ker}(f)$. We show $xh = h'x$, which is equivalent to $xhx^{-1} = h'$, for some $h' \in \mathrm{Ker}(f)$. We have $f(xhx^{-1}) = f(x)f(h)f(x^{-1}) = f(x)e_{G'}f(x)^{-1} = f(x)f(x)^{-1} = e_{G'}$. Hence $xhx^{-1} = h' \in \mathrm{Ker}(f) = H$ which implies $xh = xhx^{-1}x = h'x$. Similarly one can show $\mathrm{Ker}(f)x \subset x\,\mathrm{Ker}(f)$. □
4. (Prove the following as an exercise) A subgroup H of G is normal if and only if
$$xHx^{-1} = \{xhx^{-1} : h \in H\} = H, \ \forall x \in G.$$
   Indeed this follows easily from the fact that
$$xh = h'x \iff xhx^{-1} = h'.$$

Definition 2.4 (Conjugate Elements).
Two elements g, g' of a group G are called *conjugate* if $\exists x \in G$ with

$$g' = xgx^{-1}.$$

The conjugation relation among elements of G is an equivalence relation (check this fact).

- A subgroup H of G is normal if and only if whenever H contains an element h it contains all its conjugates $\{xhx^{-1} : x \in G\}$ (prove this as an exercise).
- If G is abelian: $xhx^{-1} = xx^{-1}h = h$ and each element equals its own conjugates.

Conjugate elements in S_n

Let $\sigma \in S_n$ and write $\sigma = (a_1 \dots a_{n_1}) \dots (b_1 \dots b_{n_k})$ as a product of k disjoint cycles of lengths $n_1 \leq n_2 \leq \dots \leq n_k$. The k-tuple $(n_1, \dots, n_k)$ is called the *cycle type* of σ.

Example 2.3.
$(12)(345)(6789)$ is of type $(2, 3, 4)$ and $(12)(34)$ is of type $(2, 2)$.

Proposition 2.5.
Two permutations in S_n are conjugates if and only if they have the same cycle type.

Proof. Let $\sigma = (a_1 \dots a_{n_1}) \dots (b_1 \dots b_{n_k}) \in S_n$ be a permutation of type $(n_1, \dots, n_k)$. Let $\tau \in S_n$. We claim

$$\tau \circ \sigma \circ \tau^{-1} = (\tau(a_1) \dots \tau(a_{n_1})) \dots (\tau(b_1) \dots \tau(b_{n_k}))$$

is the decomposition of $\tau \circ \sigma \circ \tau^{-1}$ into the product of disjoint cycles, hence it also has cycle type $(n_1, \dots, n_k)$.

We first demonstrate the equation. For this, we need to show that if $\sigma(i) = j$ then

$$\tau \circ \sigma \circ \tau^{-1}(\tau(i)) = \tau(j). \qquad (*)$$

But $\tau \circ \sigma \circ \tau^{-1}(\tau(i)) = \tau(\sigma(\tau^{-1}(\tau(i)))) = \tau(\sigma(i)) = \tau(j)$.

Moreover, the cycles $(\tau(a_1) \dots \tau(a_{n_1}))$ $\dots$ $(\tau(b_1) \dots \tau(b_{n_k}))$ are disjoint since the above cycles $(a_1 \dots a_{n_1})$, $\dots$, $(b_1 \dots b_{n_k})$ are disjoint and τ is a bijection.

Conversely let $\sigma = (a_1 \dots a_{n_1}) \dots (b_1 \dots b_{n_k})$ and $\sigma' = (a'_1 \dots a'_{n_1}) \dots (b'_1 \dots b'_{n_k})$ be two permutations with the same cycle type. Let $\alpha_1, \alpha_2, \dots, \alpha_t$ and $\beta_1, \beta_2 \dots, \beta_t$ the the elements of $\{1, 2, \dots, n\}$ such that $\sigma(\alpha_i) = \alpha_i$, $\sigma'(\beta_i) = \beta_i$, respectively, $1 \leq i \leq t$. Define the permutation $\tau \in S_n$ by $\tau(a_i) = a'_i, \dots, \tau(b_i) = b'_i$ and $\tau(\alpha_i) = \beta_i$ (convince yourself that this is a bijection). By the above property $(*)$ we have:

$$\tau \circ \sigma \circ \tau^{-1} = (\tau(a_1) \dots \tau(a_{n_1})) \dots (\tau(b_1) \dots \tau(b_{n_k})) = (a'_1 \dots a'_{n_1}) \dots (b'_1 \dots b'_{n_k}) = \sigma'$$

(check this fact) and σ and σ' are conjugate. □

Example 2.4.

1. Elements of S_n may be conjugate in various distinct ways: let $\sigma_1 = (12)(345)$ and $\sigma_2 = (34)(125)$ then $\sigma_2 = \tau \circ \sigma_1 \circ \tau^{-1}$ where $\tau = (13)(24)$ (see above proof). If we rewrite $\sigma_2 = (43)(251)$ then $\sigma_2 = \tilde{\tau} \circ \sigma_1 \circ \tilde{\tau}^{-1}$ where $\tilde{\tau} = (145)(23)$. Thus σ_1 and σ_2 are conjugate by both τ and $\tilde{\tau}$.
2. In S_3 all 3 cycles are conjugate to each others: $(123) = (23)(132)(23) = (12)(132)(12) = (31)(132)(31)$. Further in S_3 all transpositions are conjugate: $(12) = (213)(23)(231) = (13)(23)(13)$, $(13) = (21)(23)(21) = (231)(23)(213)$, and $(13) = (23)(12)(23)$.
3. In S_4 all products of disjoint transpositions are conjugate: $(12)(34) = (32)[(13)(24)](32)$, $(12)(34) = (423)[(14)(23)](432)$, and $(13)(24) = (43)[(14)(23)](43)$.

4. Being conjugate in S_n doesn't mean necessarily being conjugate in A_n. Permutations $\sigma_1, \sigma_2 \in A_4$ are conjugate in A_4 if $\sigma_2 = \tau\sigma_1\tau^{-1}$ with $\tau \in A_4$. In particular σ_1 and σ_2 must have the same cycle type, but even so they might not be conjugate in A_4: $(132) = \tau(123)\tau^{-1}$ where $\tau \in \{(12), (13), (23)\}$ but
$$\{(12), (13), (23)\} \cap A_4 = \emptyset.$$
5. All product of disjoint transpositions are conjugate in A_4: $(14)(23) = (132)[(13)(24)](123)$, $(12)(34) = (423)[(14)(23)](432)$, $(13)(24) = (134)[(14)(23)](143)$, and
$$\{(132), (423), (134)\} \subset A_4.$$

Definition 2.5 (Quotient Group Law).
Let G be a group and H a **normal** subgroup of G. We define a composition law $(.)$ on the set of left cosets G/H by
$$(.) : G/H \times G/H \to G/H$$
$$(xH, yH) \mapsto xH.yH = xyH.$$

First, we should verify that this composition law is well-defined. That is, we must show that if $x_1H = x_2H$ and $y_1H = y_2H$, then $x_1y_1H = x_2y_2H$. From the definition of the relation R, we know, for all $z_1, z_2 \in G$, that $z_1H = z_2H$ if and only if $z_1^{-1}z_2 \in H$.

Write $x_1^{-1}x_2 = h$ and $y_1^1y_2 = h'$. Then $h \in H$ and $h' \in H$. It suffices to show that $(x_1y_1)^{-1}(x_2y_2) \in H$.

Note
$$(x_1y_1)^{-1}(x_2y_2) = (y_1^{-1}x_1^{-1})(x_2y_2) = y_1^{-1}(x_1^{-1}x_2)y_2 = y_1^{-1}hy_2 = (y_1^{-1}hy_1)(y_1^{-1}y_2) = h''h',$$
where $h'' = (y_1^{-1}hy_1 \in H$ as H is normal in G.

This is well defined since H is normal: if $x'H = xH$, and $y'H = yH$ then $x'y'H = xyH$ (see discussion before Definition 2.3).

Proposition 2.6.
*$(G/H, .)$ is a group, it is called the **quotient group** of G by H.*

Proof. Associativity $(xH.yH).zH = (xy)H.zH = (xy)zH = x(yz)H = xH.yzH = xH.(yH.zH)$. The neutral element is $e_{G/H} = e_GH = H$: $xH.e_GH = xe_GH = xH$. The inverse of xH is $x^{-1}H$: $xH.x^{-1}H = xx^{-1}H = e_GH$. □

Example 2.5.

1. If G is abelian then every subgroup H of G is normal and we can consider the quotient group G/H. In this case the quotient group G/H is abelian (check this fact).
2. Let $G = \mathbf{Z}$, $H = n\mathbf{Z}$ with $n \geq 1$. The quotient group $\mathbf{Z}/n\mathbf{Z}$ equals $\{\bar{0}, \cdots, \overline{n-1}\}$, $\bar{i} = i + n\mathbf{Z}$. Its group law $(.)$ is denoted $+$ and is given by
$$\bar{i} + \bar{j} = (i + n\mathbf{Z}) + (j + n\mathbf{Z}) = (i + j) + n\mathbf{Z} = \overline{i + j}$$
this is the group $(\mathbf{Z}_n, +)$. From now on we will systematically use the notation $\mathbf{Z}/n\mathbf{Z}$ instead of $\mathbf{Z}_n$, as well as the notation $\bar{i} = i + n\mathbf{Z}$ instead of $[i]_n$.

3. $A_3 = \{e, (123), (132)\}$ is a normal subgroup of S_3, it has 2 left cosets:

$$eA_3 = A_3 = (123)A_3 = (132)A_3$$

and

$$(12)A_3 = (13)A_3 = (23)A_3 = \{(12), (23), (13)\}.$$

Thus $S_3/A_3 = \{eA_3, (12)A_3\}$ and the group law is given by: $eA_3.(12)A_3 = (12)A_3$, $eA_3.eA_3 = eA_3$, $(12)A_3.(12)A_3 = (12)^2A_3 = eA_3$. The group S_3/A_3 is an abelian cyclic group of order 2 generated by $(12)A_3$.

4. In $D_8 = \{1, r, r^2, r^3, s, sr, sr^2, sr^3\}$ the subgroup $H = \{1, r^2, s, sr^2\}$ is a normal subgroup. Indeed if i is odd

$$s^i r^j r^2 (s^i r^j)^{-1} = s^i r^{j+2-j} s^{-i} = s^i r^2 s^{-i} = r^2 \in H$$

$$s^i r^j s (s^i r^j)^{-1} = s^{i+1} r^{-j-j} s^{-i} = s^{i+1-i} r^{2j} = sr^{2j} \in H$$

and

$$s^i r^j sr^2 (s^i r^j)^{-1} = s^{i+1} r^{2-2j} s^{-i} = s^{2i+1} r^{2j-2} = sr^{2(j-1)} \in H$$

(check by yourself what happens if i is even in which case $s^i = 1$) . The subgroup H has 2 left cosets: $H = 1H = sH = r^2H = sr^2H$ and $rH = r^3H = srH = sr^3H = \{r, r^3, sr, sr^3\}$. Thus $D_8/H = \{1H, rH\}$ is a cyclic group of order 2: $1H.1H = 1H$, $1H.rH = rH$, and $rH.rH = r^2H = 1H$.

Map onto the quotient group. Let H be a **normal** subgroup of G. Consider the map

$$\phi : G \to G/H,$$

defined by

$$g \mapsto \phi(g) = gH.$$

Proposition 2.7.
The map ϕ is a surjective group homomorphism and $\mathrm{Ker}(\phi) = H$.

Proof. The fact that ϕ is surjective is clear $gH = \phi(g)$ for every $g \in G$. It is a homomorphism since

$$\phi(g_1 g_2) = g_1 g_2 H = (g_1 H).(g_2 H) = \phi(g_1).\phi(g_2).$$

We show $\mathrm{Ker}(\phi) = H$. First $H \subset \mathrm{Ker}(\phi)$ since if $g \in H$ then $e_G^{-1} g = g \in H$ and $e_G R g$, hence $\phi(g) = gH = e_G H = e_{G/H}$. Conversely let $g \in \mathrm{Ker}(\phi)$ meaning $\phi(g) = gH = e_G H$, then $e_G R g$ and $e_G^{-1} g = g \in H$. □

Going back to group homomorphisms, we will now state and prove the first isomorphism theorem for groups. Let

$$f : G \to H$$

be a group homomorphism.

The kernel $\mathrm{Ker}(f)$ of f is a normal subgroup of G [see Examples 2.2] and we can consider the quotient group $G/\mathrm{Ker}(f)$. The image $\mathrm{Im}(f) = \{h \in H : h = f(g), g \in G\}$ is a subgroup of H (see discussion after Definition 1.7).

Theorem 2.8 (First Isomorphism Theorem)**.**
The quotient group $G/\operatorname{Ker}(f)$ is isomorphic to $Im(f)$.

Proof. Consider the map

$$\pi : G/\operatorname{Ker}(f) \to \operatorname{Im}(f),$$

defined by

$$\pi(g\operatorname{Ker}(f)) = f(g).$$

We show π is a group isomorphism. First we check that π is well defined. Assume $g\operatorname{Ker}(\pi) = g'\operatorname{Ker}(\pi)$ meaning $g'^{-1}g = \tilde{g} \in \operatorname{Ker}(\pi)$. Then $f(g) = f(g'\tilde{g}) = f(g')f(\tilde{g}) = f(g')e_H = f(g')$ since $\tilde{g} \in \operatorname{Ker}(f)$. Further π is a homomorphism:

$$\pi(g\operatorname{Ker}(f).g'\operatorname{Ker}(f)) = \pi(gg'\operatorname{Ker}(f)) = f(gg') = f(g)f(g') = \pi(g\operatorname{Ker}(f))\pi(g'\operatorname{Ker}(f)).$$

The homomorphism π is surjective: if $f(g) \in \operatorname{Im}(f),\ g \in G$, then

$$f(g) = \pi(g\operatorname{Ker} f)$$

by the definition of π. Finally π is injective: assume $f(g) = \pi(g\operatorname{Ker}(f)) = \pi(g'\operatorname{Ker}(f)) = f(g')$, then $f(g')^{-1}f(g) = f(g'^{-1}g) = e_H$ and $g'^{-1}g \in \operatorname{Ker}(f)$, or equivalently gRg' meaning $g\operatorname{Ker} f = g'\operatorname{Ker} f$. □

Corollary 2.9.
Let G, H be groups with G finite and $f : G \to H$ a group homomorphism. Then $Im(f)$ is finite and

$$|G| = |\operatorname{Ker}(f)||Im(f)|.$$

In particular both $|\operatorname{Ker}(f)|$ and $|Im(f)|$ divide $|G|$.

Proof. The First Isomorphism Theorem states that $G/\operatorname{Ker}(f)$ is isomorphic to $\operatorname{Im}(f)$. Since G is finite, $G/\operatorname{Ker}(f)$ is finite, and thus $\operatorname{Im}(f)$ is finite being isomorphic to $G/\operatorname{Ker}(f)$. Further this isomorphism implies $|G/\operatorname{Ker}(f)| = |\operatorname{Im}(f)|$. On the other hand

$$|G/\operatorname{Ker}(f)| = |G : \operatorname{Ker}(f)| = \frac{|G|}{|\operatorname{Ker}(f)|}$$

by Lagrange's theorem hence the result. □

Note that if $f : G \to H$ is a surjective group homomorphism then $\operatorname{Im}(f) = H$ and the FIT for groups states that $G/\operatorname{Ker}(f)$ is isomorphic to H.

Example 2.6.

1. Consider the map $f : D_6 \to \mathbf{Z}/2\mathbf{Z}$ defined by

   $$f(s^i r^j) = \bar{i}$$

   thus $f(1) = f(r) = f(r^2) = \bar{0}$ and $f(s) = f(sr) = f(sr^2) = \bar{1}$. Then f is surjective and is a homomorphism [see Examples 1.6]:

   $$f(s^i r^j s^t r^e) = f(s^i s^t r^{-j} r^e) = f(s^{i+t} r^{i-e}) = \overline{i+t} = \bar{i} + \bar{t} = f(s^i r^j) + f(s^t r^e).$$

   Further $\operatorname{Ker}(f) = H = \langle r \rangle$ is the subgroup generated by r. The First Isomorphism Theorem asserts that D_6/H is isomorphic to $\operatorname{Im}(f) = \mathbf{Z}/2\mathbf{Z}$.

2. Find, if any, a surjective homomorphism from S_3 to a group of order 2 or 3.

- A surjective homomorphism from S_3 to a group of order 2 would have as kernel a normal subgroup of S_3 of order 3. The subgroup A_3 is a normal subgroup of order 3, and a group of order 2 must be cyclic. We construct a surjective homomorphism

$$f : S_3 \to \mathbf{Z}/2\mathbf{Z} = \{\bar{0}, \bar{1}\}$$

as follows: $f(\sigma) = \bar{0}$ is σ is even and $f(\sigma) = \bar{1}$ if σ is odd, thus

$$f(e) = f((123)) = f((132)) = \bar{0}$$

and

$$f((12)) = f((13)) = f((23)) = \bar{1}$$

(check f is a homomorphism). Further $\mathrm{Ker}(f) = A_3$ and the FIT asserts that S_3/A_3 is isomorphic to $\mathrm{Im}(f) = \mathbf{Z}/2\mathbf{Z}$.

- A surjective homomorphism from S_3 to a group of order 3 would have a kernel a normal subgroup of S_3 of order 2. Such a subgroup would contain an element of order 2, namely a transposition, hence would contain all transpositions as they are all conjugated in S_3 (see Proposition 2.5). Since there are 3 transpositions in S_3 we see that no such a subgroup, hence no such homomorphism, exists.